Chapter 4

Localization

Localization is the art of estimating the site of the maximum potential (origin of a waveform on the cortex). This depends on the montage in use and the rule of polarity, that is, the pointer deflects toward the relatively negative electrode.

Using different montages, the reader can roughly estimate the location of a waveform but unlike imaging, this technique is presumptive and imprecise. It assumes that the cortex is a smooth sphere with all its potentials arranged in the form of radial spokes. Each potential is a dipole with a superficial end that lends itself recordable from a surface electrode and a deep end that is not recordable. Of course, we know that the cortex is neither smooth nor spherical and therefore occult potentials often exist over sites such as the depths of a sulcus, inferior frontal lobes or insula that may not be detected by scalp EEG [1].

Localization Using a Bipolar Montage

Phase reversal is the key to localizing a waveform in a bipolar montage. This is the simultaneous but opposite deflection of the waveform in two channels that contain a common electrode. It implies that the potential is maximal at or near the location of the common electrode. If a longitudinal bipolar montage is in use, the next step would be to confirm the location of the phase reversal using another bipolar montage oriented perpendicular to the initial montage such as a transverse bipolar montage. Most phase reversals are negative (><), though positive phase reversals (<>) may also occur (especially in children). As mentioned previously, bipolar montages are best suited for detecting focal and not broad or generalized potentials [2]. Figure 4.1 explains the formation of a phase reversal in longitudinal bipolar montage over a hypothetical potential.

Localization Using a Referential Montage

The greatest amplitude of the deflection is key to localizing a waveform in a referential montage. This implies that the potential is maximal at or near the location of the active electrode in the channel with the maximal deflection. As mentioned previously, referential montages are best suited for detecting broad

https://doi.org/10.1017/9781108918923.006 Published online by Cambridge University Press

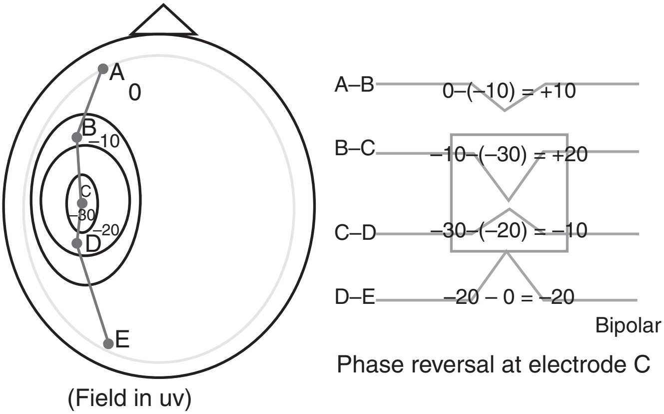

Figure 4.1 A phase reversal occurs between channels B–C and C–D as electrode C lies closest to the point of maximal potential (–30 uV).

or generalized potentials. They are limited by artifact and situations with an active reference (Chapter 5) [2]. Figure 4.2 explains the formation of maximal amplitude in a referential montage over a hypothetical potential.

Localization in Practice

A focal discharge is an epileptiform waveform that suggests the possible location of cortical irritability or tendency to generate seizures. Figure 4.3 shows a focal discharge on a longitudinal bipolar montage. There is a distinct phase reversal over the F4 electrode (channels Fp2-F4 and F4-C4). This location remains consistent when viewed in transverse montage (Figure 4.4). Furthermore, using an ipsilateral ear referential montage (Figure 4.5) shows that the discharge appears tallest (maximal amplitude) in the F4-A2 channel. Therefore, this potential focus resulting in the epileptic discharge localizes to the right frontal region (F4 electrode). As previously mentioned, this inference doesn't precisely correlate with the cortical anatomy (i.e., the right frontal lobe). It only estimates the general region where the focus of maximal potential generating the waveform of interest may originate.

Special Situations with Bipolar Montages

Isopotentials: If the maximal potential is large enough to involve two adjacent electrodes, then that channel will be neutral. There may be a phase reversal in

https://doi.org/10.1017/9781108918923.006 Published online by Cambridge University Press

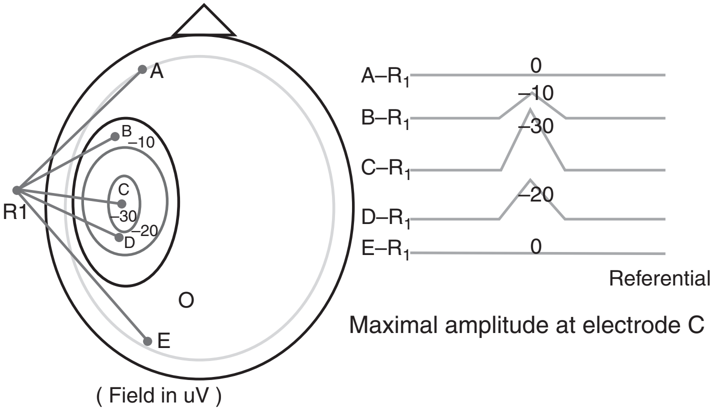

Figure 4.2 The same potential in Figure 4.1 on an ipsilateral ear referential montage. Electrode C lies closest to the point of maximal potential (–30 uV), therefore amplitude of deflection is greatest in channel C-R1.

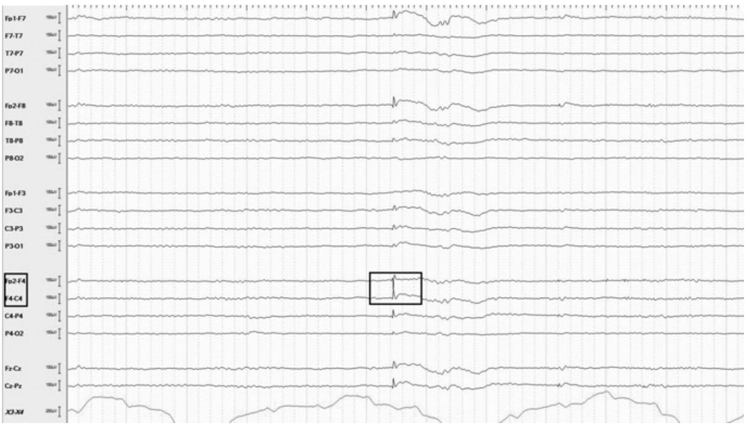

Figure 4.3 Twenty-nine-year-old man with epilepsy, discharge shows phase reversal at F4 (between channels Fp2-F4 and F4-C4) in longitudinal bipolar montage.

https://doi.org/10.1017/9781108918923.006 Published online by Cambridge University Press

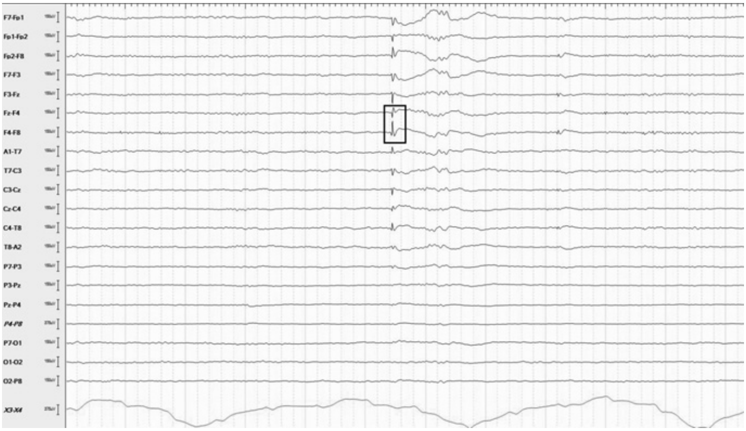

Figure 4.4 Same patient as in Figure 4.3, discharge again shows a phase reversal at F4 (between channels Fz-F4 and F4-F8) in transverse bipolar montage.

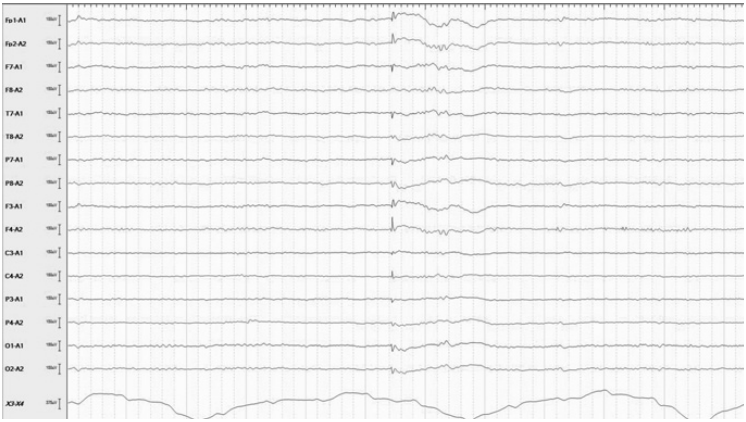

Figure 4.5 Same patient as in Figure 4.3, discharge again shows maximal amplitude at F4 (channel F4-A2) in ipsilateral ear (A2) referential montage.

https://doi.org/10.1017/9781108918923.006 Published online by Cambridge University Press

the channels above and below the neutral channel. Therefore, phase reversals need not be adjacent to each other. However, nonadjacent phase reversals are separated by an isopotential channel (nearly flat line). Figure 4.6 shows T7-P7 at isopotential in a patient with left temporal discharges.

End of Chain Potential: If the maximal potential lies at or near the last electrode (end of chain) in a bipolar montage, such as O1 in a longitudinal bipolar montage, then there will be no phase reversal as there is no electrode to straddle the potential. The waveform generated by this potential will simply have maximal amplitude at the terminal electrode without a phase reversal. In these situations, the reader should use another bipolar montage perpendicular to the initial chain (such as a circumferential montage). A referential montage can also be used.

Phase Reversals on a Referential Montage

Phase reversals are the key to localization on bipolar montages but not referential montages except in two situations.

Horizontal Dipole: If the dipole lies horizontally (not radially as is typical) then both polar ends can be recorded off the scalp leading to a phase reversal on the referential montage. These are characteristically seen in benign epilepsy with centrotemporal spikes (described in Chapter 21).

Active Reference: If the reference is active due to contamination from an artifact potential, it may also lead to an apparent phase reversal at an electrode near another distant potential. Chapter 5 describes the active reference.

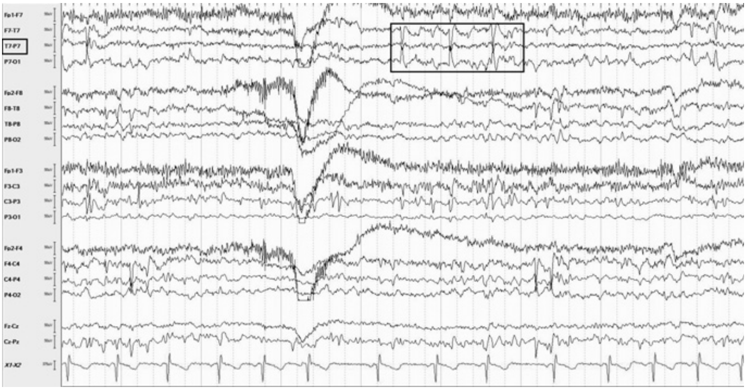

Figure 4.6 Six-year-old boy with cerebral palsy and multifocal epilepsy, left temporal discharges that show isopotential between T7 and P7.

https://doi.org/10.1017/9781108918923.006 Published online by Cambridge University Press

Chapter Summary

1. Localization is the art of locating the site of maximal potential, presumably the origin of a waveform of interest.
2. The EEG is relatively blind to certain cortical regions, such as the sulcal depths, inferior surfaces of the brain, and infoldings such as the insula.
3. Localization depends on the montage in use and the rule of polarity.
4. Phase reversal is the key to localization with a bipolar montage. It is not diagnostic of an abnormality.
5. Maximal amplitude is the key to localization with a referential montage.
6. Phase reversals are usually but not always adjacent. Nonadjacent phase reversals are separated by an isopotential channel (near flat line).
7. Foci located at the end of a chain in a bipolar montage don't show a phase reversal.
8. An apparent phase reversal on a referential montage may indicate a horizontal dipole or an active reference.

References

1. Lagerlund TD. Manipulating the magic of digital EEG: montage reformatting and filtering. *American Journal of Electroneurodiagnostic Technology*. 2000 Jun 1;**40**(2):121–36.

2. Acharya JN, Acharya VJ. Overview of EEG montages and principles of localization. *Journal of Clinical Neurophysiology*. 2019 Sep 1;**36**(5):325–9.

https://doi.org/10.1017/9781108918923.006 Published online by Cambridge University Press